

NATIONAL COLLEGE OF IRELAND: SCHOOL OF COMPUTING

MSc in Data Analytics: Research Practicum (MSCDAD_A_JAN26I)
WEEKLY PROGRESS REPORT: WEEK 1

Student Name:	Rohitkumar Amritlal Jaiswal (ID: x25119613)
Supervisor:	Dr. Thanos Staikopoulos
Project Title:	Coupling Port Congestion & Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation
Reporting Period / Date:	Week 1 (Technical Kickoff & Baseline Simulation) Tuesday, 29 September 2026

1. Summary of Activities Completed This Week

- Code Repository Setup:** Initialized the project Git repository and created a clean Python package structure named `irishlogix` using `pyproject.toml`.
- Public Data Ingestion (`cso_loader.py`):** Connected directly to Central Statistics Office (CSO) PxStat API and downloaded quarterly datasets from 2017 to 2026: TBQ01 (444 vessel arrivals), TBQ04 (2,331 Ro-Ro trailer units), and TBQ05 (3,600 trade region records). Built clean activity matrix in `data/processed/cso_port_activity_matrix.csv`.
- Jupyter Notebook Development:** Created and ran `notebooks/01_week1_data_ingestion_and_exploration.ipynb` which downloads data, plots arrival trends, shows post-Brexit trade shifts, and runs the baseline route simulation.
- Irish Road Corridor Network (`corridor_network.py`):** Built a road network graph in NetworkX with 11 nodes (Dublin, Cork, Rosslare, Athlone, Limerick, Galway, Waterford, Dundalk) and 15 road corridors with travel distances, speed limits, and tolls.
- Baseline Dispatch Simulation (`baseline_dispatcher.py`):** Built an empirical baseline simulation representing how small Irish hauliers plan routes using simple spreadsheets without delay forecasting. Ran a test of 50 freight trips: total distance 8,636.5 km; active driving time 104.4 hours; **idle waiting time 63.5 hours** (trucks wasted an average of **1.27 hours idling per trip** at port gates and customs); total diesel used 2,879.3 Litres; carbon emissions 7,601.3 kg CO2e; total transport cost €14,791.21 (average €295.82 per trip).
- Unit Testing:** Created 9 unit tests across data loader, network graph, and simulator. All 9 tests passed.

2. Primary Public Dataset Sources & URLs

CSO Table TBQ01	https://data.cso.ie/table/TBQ01	Ship arrivals for Dublin, Cork, and Rosslare ports
CSO Table TBQ02	https://data.cso.ie/table/TBQ02	Cargo tonnage categories (Ro-Ro and Lo-Lo)
CSO Table TBQ04	https://data.cso.ie/table/TBQ04	Freight trailer counts across Irish ports
CSO Table TBQ05	https://data.cso.ie/table/TBQ05	Post-Brexit trade volumes: Great Britain vs. EU
EMODnet Vessel Density	https://emodnet.ec.europa.eu/en/human-activities	AIS marine traffic density maps
OpenStreetMap Ireland	https://download.geofabrik.de/europe/ireland-and-northern-ireland.html	Road distances and speeds for Irish corridors

3. Reflection and Key Findings

- The Problem is Real:** The baseline simulation shows that without delay prediction, trucks waste an average of 1.27 hours per shipment waiting at gates and customs. This proves that coupling delay prediction with route optimization can save real fuel, money, and emissions.
- Data is Verified:** The CSO PxStat API provides clean, continuous historical data from 2017 to 2026, ensuring we have solid numbers to train the machine learning models.
- Benchmark is Set:** Having this baseline simulator running in Week 1 gives us a clear benchmark to compare our future machine learning models against.

4. Planned Activities for Next Week

- Review Week 1 baseline results and refined project plan with Dr. Thanos Staikopoulos during our meeting.
- Ingest EMODnet AIS vessel density data for Irish coastal shipping corridors.
- Start feature engineering for the Component 1 Port Congestion Classifier.
- Begin parameter calibration for the synthetic customs clearance delay generator.

5. Milestone Tracking

M1	Week 3	Ahead of Schedule	Baseline simulator working, CSO data ingested, road network built
M2	Week 9	Planned	Train XGBoost delay models and connect to Pareto optimizer
M3	Week 13	Planned	Complete comparison against baseline and build Streamlit dashboard
M4	Week 15	Planned	Final dissertation, code repository, and viva presentation

Student Signature: Rohitkumar Amritlal Jaiswal Date: 29 September 2026
Supervisor Signature: _____ Date: _____